

3 Ordering of clitics: Cartographic evidence for a universal functional sequence

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3.1 Introduction

Cartographic studies have focused on the sequences of functional elements that characterise the fine structure of the different zones of clauses and phrases, thereby reflecting a universal functional sequence (fseq). Foundational studies of the left periphery of the clause (Rizzi, 1997), the IP internal system of inflectional heads and the hierarchy of adverbs (Cinque, 1999), as well as the low zones of the clause surrounding the VP area, sometimes referred to as the VP periphery (Belletti, 2004), converge on the conclusion that rigid linear order reflects rigid hierarchical organisation. Within this framework, cross-linguistically stable ordering restrictions are taken to provide direct evidence for the existence and relative position of functional projections in the universal functional sequence (fseq).

Despite this progress, the functional space between CP and TP, traditionally treated as a landing zone for clitics, remains underexplored within the cartographic tradition compared to other parts of the functional sequence. To our knowledge, the only explicit cartographic treatment of this domain based on clitic data is the analysis of Romanian clitics in Ciucivara (2009).¹ This is striking, because clitics, whether taken individually or as components of larger clitic clusters, display clear and highly systematic placement and ordering restrictions, making them a particularly suitable object of cartographic investigation. More generally, whenever rigid ordering and placement restrictions are identified elsewhere in syntax, cartographic research takes them to provide evidence for the sequence of functional projections in the universal fseq. We argue that the same reasoning should apply to clitics as well.

In this chapter, we adopt precisely this perspective and treat the internal ordering of clitic clusters as a diagnostic for reconstructing the functional sequence between CP and TP. If the fseq is taken to be the only source of ordering, then a single clitic projection cannot be sufficient to account for the order of elements within a clitic cluster. Such an analysis would require an additional, clitic-specific source of ordering. If, however, we aim to minimise independent ordering mechanisms and take the fseq to determine order throughout the clause, it is more consistent from a cartographic perspective to assume that clitic clusters correspond to multiple projections in this part of the structure, rather than to a single clitic projection with special internal ordering rules. We therefore

¹Closely related proposals can also be found in Bianchi (2006) and Poletto (2000), both working on different Italian dialects.

ask which morphosyntactic features must be projected in this domain if the observed ordering restrictions are taken seriously.

On the basis of the cross-linguistic patterns surveyed in the following sections, we argue that clitic clusters make it possible to reconstruct the portion of the universal fseq between CP and TP, given in (1). The discussion that follows progressively motivates this hierarchy through a systematic comparison of clitic systems across languages.

- (1) CP > MoodP > New/Given_InfoP > 1/2P > DatP > AccP > TP > AspP

The argument proceeds in stages. We begin with clitic systems involving exclusively verbal or clausal features and show that their ordering reflects the independently established clausal backbone (CP > TP > AspP). We then turn to systems in which grammatical functions, person features, and case features determine the internal order of clitic clusters. By comparing increasingly complex configurations across a broad range of languages and varieties, we identify stable cross-linguistic generalisations that allow us to refine the internal structure of the clitic domain step by step. Surface variation is treated as the result of an additional movement within an otherwise fixed underlying sequence.

The chapter is structured as follows. Section 3.2 motivates a cartographic approach to clitic ordering and argues against treating the clitic domain as a single projection. Section 3.3 surveys cross-linguistic patterns in clitic clusters, proceeding from systems involving exclusively verbal features to those involving grammatical functions, person, and case, which culminates in Romance, Greek, and Slavic languages. Section 3.3.7 then compares Czech clitic clusters with the resulting pattern. Section 3.3.8 synthesises the generalisations emerging from the survey and refines the proposed hierarchy of functional projections. Section 3.4 discusses the further theoretical implications of the analysis for deriving clitic ordering from the universal functional sequence. Section 3.5 concludes the chapter.

3.2 Rigid clitic ordering calls for a cartographic approach

In current syntactic theory, it is standard to assume that there is a functional domain situated between CP and TP, often labeled as the clitic zone.² While it is well established that clitics occupy this space, there is little consensus on which syntactic features are actually encoded within

²We are, of course, aware that there are two main approaches to the representation of the clitic zone. One assumes a sequence of distinct, rigidly ordered clitic heads (e.g., Haegeman, 1993, Bošković, 2004, Tanenbaum, 2020, Tabachnick, 2024). The other treats clitics as a single projection, often derived as an instance of Matushansky's (2006) *m-merger*, which is responsible not only for the creation of the clitic cluster, but also for forming a constituent that includes both the clitics and their host (see e.g., Terzi, 1999, Martinović, 2020, etc.). We adopt the former view, since cross-linguistic evidence shows that clitic clusters can be interrupted by adverbs or various types of particles. Rather than assuming a single clitic head, we propose that clitics occupy a dedicated domain in which multiple features are projected separately and ordered rigidly because they follow a fixed functional sequence. This, we argue, accounts for the high degree of cross-linguistic stability in clitic ordering. For discussion of clitic cluster interruption by non-clitic elements, see Čavar and Wilder (1997) and Stjepanović (1998) for Serbo-Croatian, Billings (2005) for Tagalog, Lenertová (2004) for Czech, Bošković (2001) for Slovenian.

this clitic zone, or in what order these features are organised. As a result, the clitic zone often functions as a placeholder: a cover term for a domain whose internal structure remains unclear.

This unresolved issue can be represented schematically as in (2), where the question marks indicate that it remains unclear what occupies the structural space between CP and TP within the clitic zone, or at least that there is no consensus on how this domain is internally organised.

(2) CP > clitic zone > TP ↔ CP > ??? > TP

Despite the unresolved internal organisation of the clitic zone, one property is universally acknowledged: clitics are rigidly ordered with respect to one another. Yet rather than being taken as evidence of internal organisation, this rigidity is often treated as a problem: as a surface effect that resists syntactic analysis, precisely because no obvious, cross-linguistically shared feature organisation has been identified. As a result, the clitic zone has rarely been examined through the lens of the cartographic approach, which traditionally relies on the identification of rigid sequences.

Our proposal is to take this rigidity of clitic ordering as a clue. We suggest that this domain has an internal structure that can be made visible through cartographic analysis, despite all the challenges this approach to clitics entails. By examining ordering patterns across typologically diverse languages, we propose that a cross-linguistically stable sequence can be identified behind the clitic zone, and that it is a suitable candidate for cartographic analysis.

The scope of this chapter is limited. We do not aim to resolve all problems connected with clitics. Our goal has been to take a first step toward uncovering the structure of the clitic zone through a systematic cross-linguistic investigation. We view this as a starting point for further research and as a promising way to establish a baseline for the analysis of clitic ordering.

The main discussion of this chapter takes up this task by examining clitic data from over 30 languages and their varieties, with the aim of identifying generalisations about the structure of the universal functional sequence. Rather than analysing clitic clusters for their own sake, we treat them as empirical evidence that can reveal the internal organisation of the universal fseq, a perspective previously adopted rarely, we can find it in Bianchi (2006) or Poletto (2000). The discussion proceeds gradually: it begins with verbal clitics, whose ordering most transparently reflects the broadly recognised feature hierarchy of the clause structure. This alignment offers initial support for the approach. We then turn to clusters that reflect additional projections, allowing us to uncover previously uncharted parts of the functional sequence behind the clitic zone. The next section undertakes this investigation in detail.

3.3 Cross-linguistic patterns in clitic ordering

There is a broad consensus in the literature that ordering restrictions reflect an underlying hierarchy of functional projections, provided one adopts the cartographic and nanosyntactic assumption that every feature projects. This assumption, known as the One Feature One Head principle (Cinque and Rizzi, 2008), taken together with the observed complexity of the clitic zone, implies that the area between CP and TP must contain a rich sequence of functional projections. Clitics encode a wide range of morphosyntactic features (Tense, Mood, Case, Person, etc.), and their rigid ordering,

required precisely in this field, makes it clear that a single projection like CIP is insufficient to account for the internal structure of the clitic zone.

The goal of this section is to survey the results of previous analyses of clitic clusters that have led to multiple generalisations about their internal ordering. Due to space restrictions and because detailed examples would lead us too far afield, this chapter does not provide specific instances of clitic clusters but focuses only on these generalisations. However, for each language discussed, we include references to the relevant literature where such examples can be found for further investigation. What we will see is that different languages order clitics according to different types of features.

The survey proceeds in steps, moving from relatively simple clitic systems to increasingly complex ones. We begin with languages where clusters contain only verbal clitics, then turn to clusters ordered by grammatical functions or case, followed by systems where person features play a role. Next, we consider languages where both person and case contribute to the ordering, and finally, we examine Slavic languages, where all these features are reflected in the cluster. This section concludes with a summary of the overall generalisations.

The following section begins with examples of clitic clusters that include only verbal and/or clausal features.

3.3.1 Verbal/clausal features

Whenever verbal clitics are involved, the clitic cluster respects the clausal functional sequence. At the top is the Complementiser Phrase (CP), which hosts complementisers (e.g., conjunctions or *wh*-expressions) and discourse-related features (e.g., topic and focus). Below CP is MoodP, which conveys the speaker's attitude or intention towards the event, ranging from certainty and possibility to obligation or desire. The next projection is Tense (T), which typically expresses future or past time reference, while present tense is markedly less common. At the bottom of the hierarchy is Aspect (AspP), which encodes the internal temporal structure of the event.

(3) CP > MoodP > TP > AspP

Languages such as Datooga and Ihanzu–Nyilamba illustrate this point (Table 3.1). Even though their clusters are relatively simple, they still conform to the predicted hierarchy, suggesting that clitic order is shaped by the universal fseq.

Table 3.1: Clitic clusters ordered by exclusively verbal/clausal features

reference	language	cluster
Harvey et al. (2023)	Datooga (Nilotic)	Cond — Polar(Aff/Neg) — T — Asp
	Ihanzu–Nyilamba (Bantu)	C _(subordinating) — Neg — T

Table 3.1 shows that the clitic cluster may include clitics belonging to the CP domain. In Ihanzu–Nylamba, clusters in subordinate clauses begin with a clitic marking subordination, followed by a negation clitic and a tense clitic. The clitic cluster thus closely matches the functional sequence generally assumed for the higher portion of the clause structure, as shown in (3).

Besides complementiser clitics, there may also be clitics marking polarity, indicating either affirmation or negation. We do not address the exact positioning of negation here. As shown by De Clercq (2013) and De Clercq and Vanden Wyngaerd (2019), Neg can project at several points in the clause and take on different representations. This implies that the placement of Neg within clitic clusters may vary across languages, depending on the type of negation employed. A full treatment of this issue lies beyond the scope of this chapter.

When we look at the other end of the clitic cluster, we find that clitic clusters can also draw on lower domains, such as Aspect, most commonly marking perfectivity. The data, therefore, confirm that the order of verbal clitics reflects the underlying verbal/clausal functional sequence. This constitutes the first piece of evidence that clitic clusters reflect the ordering of the universal fseq as known from clause structure in (3). Subsequently, we will see clitic clusters that continue to respect this sequence, while being enriched with additional projections. These additional projections provide further evidence about which features belong to the domain here referred to as the clitic zone.

In the next section, we turn to clitic clusters that, in addition to verbal features, also reflect grammatical functions, case and person, that is, nominal properties, and examine how these are ordered relative to one another.

3.3.2 Verbal and grammatical function features

In addition to verbal clitics, many languages also have clitics encoding the grammatical functions³ of subject and object. This raises the possibility of predicting the position of subject and object within the verbal functional sequence based on the ordering of such clitic clusters.

The languages represented in Table 3.2 lack overt case marking, so the only indicator of the grammatical function carried by a pronominal clitic is its position in the cluster. We have included examples of such clusters to show that grammatical functions are reflected in the clitic cluster even in languages without overt case marking. This is crucial, since the claim we aim to advance about the universal functional sequence is intended to be truly universal, applying across languages, including those without overt case, and at least to all the languages examined in this study. Since we do not know exactly which features are involved beyond establishing subject and object roles in these languages, we will, for the sake of simplicity, refer to the projections associated with these grammatical functions by the functions themselves, that is, as subject (Subj) and object (Obj).

³We use the term grammatical functions in line with the cited literature, which argues that neither case nor thematic role adequately captures what appears in subject and object positions. The more general label grammatical function is therefore preferable, as it is broad enough to cover the full range of functions that clitics may realise in these positions. In this sense, subject and object are understood here in the broadest possible way, without further feature specification. This is particularly relevant for languages in which the very existence or status of case is itself debated.

Table 3.2: Clitic clusters ordered by both grammatical functions and verbal features

reference	language	cluster
Harvey et al. (2023)	Iraqw–Gorwaa (Bantu)	$C_{(subordination)}$ — Mood — Voice — Subj — Obj — Asp
	Alagwa (Cushitic)	$C_{(subordination)}$ — Mood — Subj — Obj — T
	Nyaturu (Bantu)	$C_{(subordination)}$ — Mood — Subj — Obj — T

Table 3.2 shows that in all three languages, clitic clusters begin with a subordinating complementiser clitic, followed by a mood-related clitic, a subject clitic, and an object clitic. Iraqw–Gorwaa additionally includes a dedicated voice clitic between mood and subject, and its cluster ends with an aspect clitic. In contrast, the clusters in Alagwa and Nyaturu end with a tense clitic. These patterns reflect ordering based on the verbal (clausal) hierarchy and indicate the structural positions of subject and object clitics within it.

The crucial point, then, is that whatever features underlie the assignment of grammatical functions in such languages, they are relevant to the clitic cluster, as they give rise to the rigid ordering observed within it. Following the cartographic methodology, which assumes that rigid ordering reflects the order in which features are merged, we conclude that these features must be projected in precisely this order. Based on the data in Table 3.2, we thus conclude that the universal fseq is as shown in (4).

- (4) CP > MoodP > SubjP > ObjP > TP > AspP

Table 3.2 shows that both grammatical functions, subject and object, are positioned above the temporal projection T, however, at least one language deviates from this pattern, as shown in the next table.

Table 3.3: Clitic clusters ordered by both grammatical function and verbal features (troublemaker)

reference	language	cluster
Barrett-Keach (1986) Deen (2013)	Swahili (Bantu)	(Neg) — Subj — T — Obj

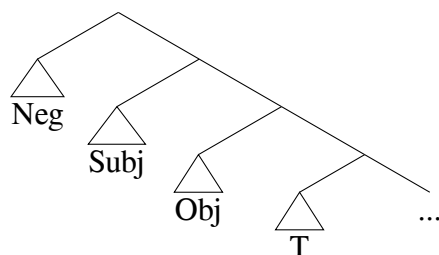
At first sight, Swahili might appear to be a counterexample to our conclusion in (4). Some linguists indeed adopt this view and interpret such cases as evidence that clitic clusters are formed in at least two distinct domains: one v/V-external and another v/V-internal, a view explicitly argued for by Cardinaletti (2008). This analysis aligns with the minimalist assumption that the object role is assigned in the v/V-internal domain, while the subject role is assigned in the v/V-external domain,

in this case identified with the overtly marked Tense. This raises the question of how such surface orders (Table 3.2 vs Table 3.3) are to be reconciled with a universal base structure, which brings us to a broader point about how the cartographic approach accounts for variation.

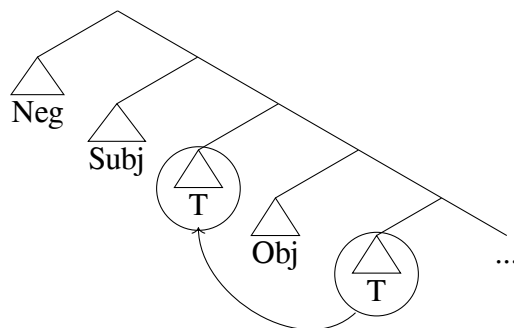
The cartographic methodology is not based on the assumption that surface order straightforwardly mirrors the universal functional sequence. As cross-linguistic data clearly show, not all features appear in the same linear order across languages. While rigid ordering can directly reflect the universal fseq, it can also do so indirectly. As argued by Cinque (2005) and Cinque and Rizzi (2008), certain surface orders can be derived from a universal base structure through syntactic movement. Crucially, however, not all permutations are possible: if we assume a fixed underlying sequence, movement operations are subject to restrictions, and only a subset of logically possible surface orders can be generated.

This insight is directly relevant for Swahili, whose clitic cluster appears to be shaped both by the universal fseq and by movement operations permitted under the restrictions identified by Cinque and Rizzi. We assume that the base order corresponds to Table 3.2, and that the surface order derived through movement is shown in Table 3.3. As also suggested by Cardinaletti (2008), Swahili behaves differently from most other languages. However, unlike Cardinaletti, we do not interpret this as evidence that Swahili reveals an underlying structure hidden elsewhere. On the contrary, we argue that Swahili masks a structure that other languages display more directly.

- (5) a. Iraqw-Gorwaa, Alagwa, Nyaturu



- b. Swahili



This analysis is illustrated in (5). Structure (5a) represents the base configuration, in which clitics are merged in the order determined by the universal functional sequence. Structure (5b) shows the derived surface order, where the T clitic undergoes movement to a position preceding the object

clitic. This derives the attested order in Swahili while preserving the underlying hierarchy assumed in Table 3.2.

What is crucial, then, is that in the base configuration, (5a), the temporal projection is merged below the object projection, following the universal sequence NegP > SubjP > ObjP > TP. In the derived structure, (5b), however, T undergoes movement to a higher position, resulting in the surface order NegP – SubjP – TP – ObjP. This leads to the following generalisation, see (6).

- (6) a. generally: Grammatical_functionP > TP
 b. in more detail: SubjP > ObjP > TP

Table 1.2 also illustrates that the grammatical functions in question are realised below Mood. This observation motivates the generalisation in (7).

- (7) a. generally: MoodP > Grammatical_functionP
 b. in more detail: MoodP > SubjP > ObjP

Together, these generalisations place the subject and object projections between the Mood and T projections within the clitic cluster. We therefore conclude that the universal hierarchy is as shown in (8).

- (8) a. generally: MoodP > Grammatical_functionP > TP > AspP
 b. in more detail: MoodP > SubjP > ObjP > TP > AspP

To sum up, we have seen that grammatical functions such as subject and object can determine clitic order, even in the absence of overt case marking. In the following section, we turn to clitic clusters in which person is the determining factor. This will allow us, first, to establish that the person plays an independent role in the clitic zone and is therefore projected within the higher functional field. Second, it will help us examine how person features interact with grammatical function features, and begin to uncover the relative position of these features within the universal functional sequence.

3.3.3 Person features

In this section, we focus on the role of person features in determining clitic order. We begin with languages in which clitics are ordered exclusively according to person. These cases demonstrate that person features alone can govern the structure of the clitic cluster, confirming that person is a relevant feature within the clitic zone. We then turn to languages where the person interacts with all the previously discussed features: verbal and clausal projections, and grammatical functions.

As we have already mentioned, in some languages, the person is the primary feature determining clitic order. In such systems, clitics follow a well-known markedness hierarchy in (9): 1ST and 2ND person features are more marked than 3RD person. As a result, clitics with 1ST and 2ND person features are placed higher in the universal functional sequence than 3RD person clitics. This is discussed, among others, by Haspelmath (2022).

(9) 1/2P > 3P

To illustrate this point, Table 1.6 presents examples of languages that have only pronominal clitics whose ordering is determined exclusively by person features. In Atayal, Warlpiri, Pashto, Squliq Atayal, and Mayrinax Ts'uli', clitics referring to 1ST and 2ND person (speech-act participants) systematically precede 3RD person clitics. These patterns show that in certain systems, the person alone can govern the internal order of the clitic cluster. The data in the table directly reflect the person markedness scale given in (9).

Table 3.4: Clitic clusters ordered exclusively by person features

reference	language	cluster
Rau (1992)	Atayal (Austronesian)	1 — 2 — 3
Legate (2008)	Warlpiri (Pama–Nyungan)	1/2 — 3
Tegey and Robson (1996) Pescarini (2021)	Pashto (Indo-European)	1 — 2 — 3
Holmer and Billings (2014)	Squliq Atayal, Mayrinax Ts'uli' (Austronesian)	1/2 — 3

These data support the conclusion that person features play a systematic role in determining the internal order of the clitic cluster. This confirms that the person is a relevant feature within the clitic zone and should therefore be projected there. To further refine the position of the person within the universal functional sequence, we now turn to more complex clitic clusters where it interacts with other features.

3.3.4 Person and grammatical function features

We now turn to a language in which clitic clusters are shaped by the interaction of two types of features: grammatical function and person. The relevant example comes from Wolof and is illustrated in Table 3.5.

Table 3.5: Clitic clusters ordered by both person features and grammatical functions

reference	language	cluster
Martinović (to appear)	Wolof (Niger–Congo)	Subj — Obj <i>and</i> Obj[1 — 2 — 3]

Table 3.5 shows that Wolof clitic clusters reflect both grammatical function and person: subject clitics always precede object clitics, regardless of the person feature they bear (10a). Among object clitics, person determines internal order: 1ST and 2ND person objects precede 3RD person ones (10b).

- (10) a. SubjP > ObjP
b. 1/2P > 3P

These data indicate that subject clitics are associated with a dedicated feature that guarantees their position above all object clitics, regardless of person features (11).

- (11) a. generally: SubjP > PersonP
b. in more detail: SubjP > 1/2P > 3P

Example (11) reflects the conclusion that subjecthood is encoded in a projection structurally higher than person projections. We tentatively label subject projection SubjP, as the Wolof data do not provide further evidence about its precise nature.

Returning now to the conclusions about the functional sequence from Section 1.3.2, repeated below as (12a), and combining them with the current findings in (11), we can further refine the structure of the universal fseq, as shown in (12b-c).

- (12) a. CP > MoodP > SubjP > **ObjP** > TP > AspP
b. CP > MoodP > SubjP > **PersonP** > TP > AspP
c. CP > MoodP > SubjP > **1/2P > 3P** > TP > AspP

However, since cross-linguistically object clitics are not ordered solely based on person features, but also according to case, the next section turns to another feature that influences clitic ordering – namely, case. We begin by examining a language in which case is claimed to be the only relevant feature for clitic ordering, thereby demonstrating that case alone can determine the structure of the clitic cluster. We then turn to systems where case interacts with other features already shown to affect clitic ordering. First, we consider clusters that combine verbal features and case, and later also more complex combinations. This leads us to the most intricate types of clitic clusters, found primarily in Romance, Slavic, and Bantu languages.

3.3.5 Case features

We now turn to the role of the case in determining clitic order. Table 3.6 presents languages in which clitic order is claimed to be determined solely by case features. Although such systems are rare cross-linguistically, they are crucial for demonstrating that case features alone can determine the internal structure of the clitic cluster. This is important because only features that contribute directly to clitic ordering within the cluster can be assumed to be part of the clitic domain.

In Kapampangan and Mamanwa (Austronesian), clitic clusters appear only in non-actor voice constructions, where genitive clitics (marking the actor/agent) always precede nominative ones (marking the undergoer/patient).⁴ Any other order is ungrammatical.

Table 3.6: Clitic clusters ordered exclusively by case features

reference	language	cluster
Holmer and Billings (2014)	Kapampangan, Mamanwa (Austronesian)	Gen(Erg) — Nom(Abs)

Languages like Kapampangan and Mamanwa thus show that case projections themselves are directly relevant to the derivation of the clitic cluster.

Since we have already established that case is relevant for determining clitic order, the next step is to examine where case projections are located within more complex clitic clusters, specifically, those that also include verbal features. We therefore turn to Kambera (Austronesian), which provides precisely this type of system. Relevant data are shown in Table 3.7.

Table 3.7: Clitic clusters ordered by both case and verbal features

reference	language	cluster
Klamer (1997)	Kambera (Austronesian)	Mood — Dat — Acc — Asp
		<i>or</i> Mood — Gen — Dat — Asp

Table 3.7 shows clitic clusters in Kambera (Austronesian) that are ordered based on both case and verbal features. When a sentence contains two object clitics, these may appear in two case combinations: either dative and accusative, or genitive and dative. In the first configuration, the dative clitic precedes the accusative; in the second, the genitive clitic precedes the dative. In both cases, the object clitics are flanked by verbal projections: Mood precedes them, and Aspect follows. This ordering matches the sequence established by the generalisations discussed so far.

Taking into account the data in Table 3.7, we arrive at the following generalisation about the position of case within the universal functional sequence, as shown in (13).

- (13) CP > MoodP > CaseP > (TP) > AspP

Unfortunately, Kambera does not have clitics expressing Tense, so we cannot directly determine the position of CaseP with respect to TP based on this language alone. However, since the

⁴This does not mean that pronominal clitics in these languages lack person features or other specifications. What we intend to show is that only case features may be relevant for the ordering of the cluster, while other features may be present but do not affect its internal arrangement.

clitics involved are object clitics, it is reasonable to assume that the case in Kambera reflects object-related grammatical functions. This allows us to infer that the Tense projection is likely located below the projections associated with grammatical functions and above the aspectual domain. This assumption aligns with the sequence in (12a), repeated here as (14a,b).

- (14) a. CP > MoodP > SubjP > ObjP > TP > AspP
 b. CP > MoodP > SubjP > CaseP > TP > AspP
 c. CP > MoodP > SubjP > PersonP > TP > AspP

Further, when we compare (13) with (12b), repeated as (14b), we arrive at the conclusion in (15). This sequence reflects the decomposition of the general object position into more fine-grained person and case projections.

- (15) CP > MoodP > SubjP > PersonP/CaseP > TP > AspP

So far, we have only examined languages whose clitic clusters are organised either by person features or by case features, but not both. As a result, the relative ordering of these two types of features remains unclear (Person/CaseP). In the following section, we turn to languages where both person and case features are active, which will allow us to investigate their mutual ordering within the functional sequence.

3.3.6 Case, person and grammatical function features

In this section, we first examine clitic clusters with object clitics only (Section 1.3.6.1), to determine whether person or case is higher in the functional sequence. These data allow us to explore how person, case, and grammatical function features interact in determining clitic order. We then turn to clusters with both subject and object clitics (Section 1.3.6.2), to uncover the nature of the Subj projection.

3.3.6.1 Pronominal clitic clusters

We now turn to languages whose clitic clusters include pronominal clitics and reflect both person and case features, allowing us to determine the relative ordering of these projections within the clitic zone.

This pattern is found in Romance languages (Spanish, Catalan, French, Italian, Romanian), in Bantu languages (Haya, Kinyarwanda, Tshiluba), and in Greek, among others. In these systems, two hierarchies interact:

- (16) a. person hierarchy: 1/2P > 3P (Haspelmath, 2022)
 b. case hierarchy: DatP > AccP (Caha, 2009 and the chapter 2 *The SE illusion: Argumental looks, antipassive reality* by Medová)

Table 3.8 presents languages in which both person and case features determine the structure of the clitic cluster. Before analysing these clusters, three points need to be noted.

First, reflexive and impersonal clitics are included in the generalisations for completeness, but we do not examine them in detail. This is due both to substantial cross-linguistic variation in their placement and to the diversity of existing analyses, which often lead to conflicting conclusions about their status and position. For these reasons, identifying their precise location within the functional sequence lies beyond the scope of this chapter.

Second, the distinction between enclitic and proclitic placement may introduce surface variation. This is especially relevant in languages like French and Greek, where the two contexts can yield different clitic orders. Where such differences are attested, the table explicitly specifies which ordering corresponds to which environment.

Third, the clusters themselves are often complex. They may vary across dialects and are frequently subject to the Person Case Constraint (PCC), which rules out certain clitic combinations. As a result, different authors report different orderings. To reflect this, some languages are listed with two or more possible patterns, linked by *and* or *or*. Our goal is not to determine which analysis is correct, but to include the broadest possible range of attested clitic orders. For this reason, some rows show multiple scales.

When scales are separated by *or*, the language allows either order, depending on context (e.g., enclisis vs proclisis in French and Greek). When they are separated by *and*, both scales are active but cannot be reduced to one unified hierarchy, largely due to PCC effects (e.g., Spanish). In such cases, scholars have identified two valid scales, both of which correctly predict the ordering of clitics, but the available evidence is insufficient to merge them into a single unified hierarchy.

From this overall variation, we aim to extract generalisations that are shared across all the systems represented.

In what follows, we will go through the individual examples in Table 3.8.

Table 3.8: Clitic clusters ordered by person and case features (pronominal clitics)

reference	language	cluster
Perlmutter (1970) Bonet (1991) Desouvrey (2005)	Spanish (Indo-European)	Refl/Impers — 2 — 1 — 3 <i>and</i> Dat — Acc <i>or</i> Refl/Impers — 2 — 1 — 3(Dat) — 3(Acc)
Monachesi (1997) Cardinaletti (2008)	Italian (Indo-European)	1/2 — 3 <i>and</i> Dat — Refl — Acc — Impers
Bonet (1991)	Standard Catalan (Indo-European)	Refl — 2 — 1 — 3(Dat) — 3(Acc)
	Valencian Catalan (Indo-European)	1 — 2 — 3(Dat) — 3(Acc)
Bonet (1991) Seuren (2009)	French (Indo-European)	proclitic: Refl/1/2 — 3(Dat) — 3(Acc) <i>or</i> enclitic: 3(Acc) — 3(Dat)/1/2
Wanner (1978) Terzi (1999) Chatzikyriakidis (2010)	Greek (Indo-European)	proclitic/enclitic: Dat — Acc <i>or</i> enclitic: Acc — Dat
Cardinaletti (2008)	Haya, Kinyarwanda, Tshiluba (Bantu)	1 — 2 — 3(Acc) — 3(Dat)

Table 3.8 shows that both hierarchies from (16) are active in Spanish. Clitics marked for 1ST and 2ND person precede those marked for 3RD person (as in (16a)), and within the 3RD person, datives precede accusatives (as in (16b)). Italian displays the same pattern: 1ST and 2ND person clitics precede 3RD person clitics, and datives come before accusatives. In Standard Catalan, 2ND and 1ST person clitics also precede 3RD person ones, which are again ordered by case. The generalisations in (16) are thus respected.

Different languages display different orderings of 1ST and 2ND person clitics. This variation can even be observed within a single language: for instance, Valencian Catalan and Standard Catalan differ in the relative ordering of these clitics, as shown in (17).

(17) 2 — 1 (Valencian) vs 1 — 2 (Standard Catalan or Spanish)

Even so, this clitic order still supports the conclusion that 1ST and 2ND person clitics (marked person) take precedence over 3RD person clitics (unmarked person), and that 3RD person clitics are ordered according to the case hierarchy. Due to the considerable variation in the ordering (and also semantics, e.g., exclusive vs inclusive meaning) of 1ST and 2ND person clitics across languages,

we set aside the question of whether one consistently precedes the other. Instead, we adopt the more general conclusion that clitics with marked person features (1ST/2ND) always precede those with unmarked person features (3RD).

In French, clitic order is also shaped by both person and case, but differs between proclitic and enclitic contexts. In proclisis, 1ST, and 2ND person clitics precede 3RD person datives, which are then followed by accusatives; in enclisis, the order is reversed, see (18):

- (18) a. proclitic: 3(Acc) — 3(Dat)/1/2
 b. enclitic: 1/2 — 3(Dat) — 3(Acc)

Crucially, in both cases, elements closer to the host are those which are expected to be merged earlier. The generalisation from (16) thus still holds.

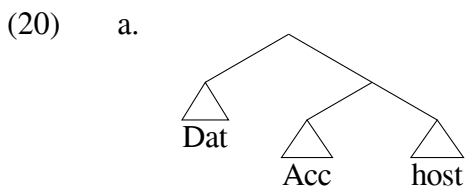
Greek clitics at first seem to pose a problem. Unlike the majority of languages discussed so far, where linearisation straightforwardly matches the underlying structure, Greek shows two surface orders that appear contradictory. They are repeated here in (19):

- (19) a. proclitic: Dat — Acc
 b. enclitic: Acc — Dat
 c. enclitic: Dat — Acc

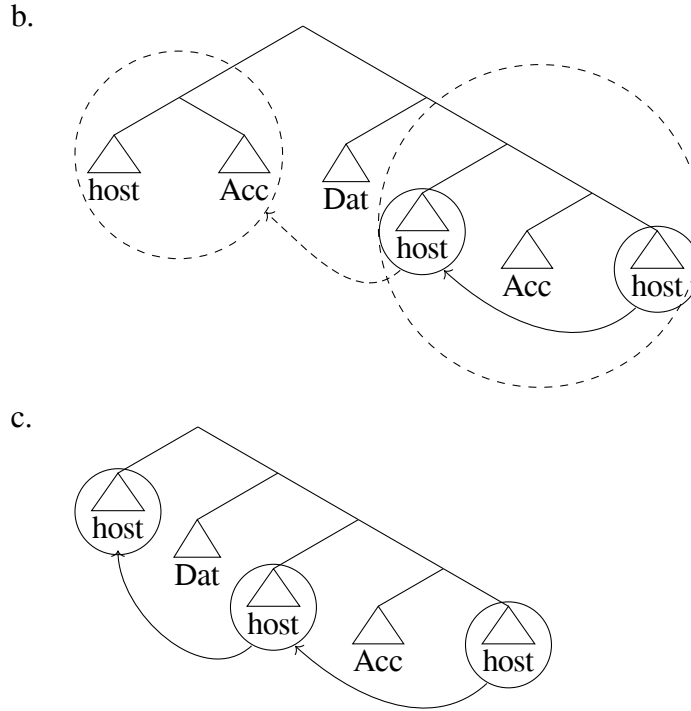
In proclitic contexts, datives precede accusatives, while in enclitic contexts, both orders are attested. As in French, the proclitic order Dat > Acc from (19a) is mirrored by an enclitic order Acc > Dat from (19b).

In the proclitic configuration (19a), the ordering is derived by simply merging first the accusative and then the dative clitic above the host, yielding the structure $[[\text{Dat}][[\text{Acc}][\text{host}]]]$ in (20a).

In enclitic configurations (19b-c), by contrast, the derivation involves additional movements. Both variants begin by merging the host with the accusative clitic, forming $[[\text{Acc}][\text{host}]]$. Subsequently, the host raises above the accusative, resulting in $[[\text{host}][\text{Acc}]]$. At this point, the dative clitic merges, creating $[[\text{Dat}][[\text{host}][\text{Acc}]]]$. To derive the order in (19b), the entire constituent $[[\text{host}][\text{Acc}]]$ moves across the dative clitic in (20b).⁵ To derive the order in (19c), only $[\text{host}]$ moves across the dative clitic in (20c).



⁵In other words, movement of the host plus pied-piping of the accusative clitic.



In (20), we have shown that if the accusative clitic merges before the dative, all three surface linearisations can be derived. Even though the surface orders look contradictory, they all originate from the same underlying hierarchy $\text{DatP} > \text{AccP}$, which is precisely the hierarchy we argue for.

A further issue is that Greek at first seems to reflect only case and not person in clitic cluster derivation. However, person distinctions are active too, as shown by restrictions on possible clitic combinations. The presence of the Person Case Constraint (PCC) in Greek, and in fact across all languages in Table 3.8, provides additional evidence for the structural role of both person and case (for the discussion of PCC see Perlmutter, 1970, Duranti, 1979, Anagnostopoulou, 1994, Cardinaletti, 2008, Cabré and Fábregas, 2019, Pescarini, 2021 etc.). These constraints show that person and case features are not just co-present in the clitic system, but interact in ways that determine which clitic orderings are grammatical. The clitic cluster is thus shaped by the structural integration of both feature hierarchies.

Table 3.8 further strengthens our hypothesis that both types of features – person and case – play a role in the formation of the clitic cluster, and that within each type the features are ordered according to the hierarchies from (16). Yet, because these two sets of features interact, it becomes crucial to establish their relative ordering as well: is person projected above case, or case above person? The languages in Table 3.8 already offer a clue, as they show that dative and accusative features matter primarily for the ordering of 3RD person clitics (or, in an alternative interpretation, clitics unmarked for person). This pattern indicates that case features play a role only as long as higher person features (1ST and 2ND) are not involved. In other words, the role of case is visible only with 3RD (unmarked) person, but is overridden once person features (1ST/2ND) come into play.

To summarise, even though Greek and French show surface variation in clitic ordering, this variation does not undermine the structural generalisation: clitics bearing marked person features (1ST/2ND) are linearised higher in the cluster than clitics that display unmarked 3RD person features and case, respecting Dat > Acc. We take this as a cartographic argument that the case projections are merged lower than person projections (1ST/2ND) in the universal functional sequence. The general pattern can thus be stated as seen in (21):

- (21) a. generally: 1/2P > 3(Case)P
 b. in more detail: 1/2P > DatP > AccP

This allows person-based projections to override lower case-based projections: case-based ordering is established first, but can be subsequently reshuffled by person-based ordering, which applies later in the derivation.

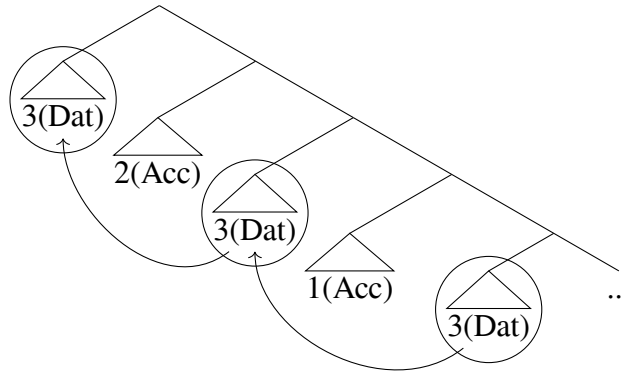
Two languages, however, appear to pose challenges to our generalisation in (21): Barceloní Catalan and Romanian (see Table 3.9).

Table 3.9: Clitic clusters ordered by person and case features (troublemakers)

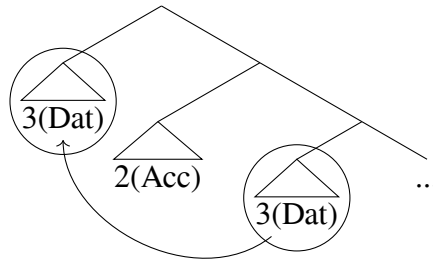
reference	language	cluster
Bonet (1991)	Barceloní Catalan (Indo-European)	2 — 1
		<i>and</i>
		3(Dat) — 2/1(Acc)
		<i>or</i> 2/1(Dat) — 3(Acc)
Ciucivara (2009)	Romanian (Indo-European)	1 — 2/3 and 3(Dat) — 2(Acc)
		<i>or</i> 1 — 2/3 and 2(Dat) — 3(Acc)

In Barceloní Catalan, a 3RD person dative precedes 1ST and 2ND person clitics, which might seem to contradict our hypothesis. Romanian exhibits similar cases, where a 3RD person dative precedes a 2ND person accusative. However, in all these cases, the disruption involves a dative clitic rather than an accusative one. This suggests that the higher-ranked case on the case hierarchy interferes with person-based ordering, likely because it is structurally closer to the person layer. Noyer (2001) describes a comparable pattern in Nunggubuyu (Pama–Nyungan family), noting that “*only peripheral clitics invert with their neighbour in the string*”. These observations support the conclusion that clitic clusters follow the fixed order of the functional sequence, and when the surface order appears to diverge from it, movement operations are responsible for the final linear arrangement. In (22a), the movement of 3RD person dative clitic is shown for Barceloní Catalan, and in (22b) for Romanian.

(22) a. Barceloní Catalan



b. Romanian



To conclude, the surface linearisations of clitic clusters in Barceloní Catalan and Romanian do not directly correspond to the fseq. This does not, however, contradict our assumption that clitic clusters reflect the universal fseq. As in other cartographic analyses, movement may obscure the original ordering of projections. The crucial point is that once movement is taken into account, the surface linearisations remain derivable from the universal fseq.⁶

So far, we have focused only on clitic clusters that include object clitics. This has allowed us to establish the relative ordering of person and case projections within the functional sequence. The general fseq is shown in (23a), and a more detailed version, distinguishing between marked person and case types, is provided in (23b).

- (23) a. generally: CP > MoodP > SubjP > PersonP > CaseP > TP > AspP
 b. in more detail: CP > MoodP > SubjP > 1/2P > DatP > AccP > TP > AspP

In the following section, we turn to clitic clusters that, in addition to object clitics, also include cliticised verbal auxiliaries. We evaluate whether these clusters likewise conform to the functional sequence in (23). If they do, they provide further support for the claim that clitic clusters across languages follow a uniform ordering that reflects the universal functional sequence.

⁶For discussion of movement typology in the cartographic framework, see Cinque (2005), Starke (2009), Baunaz et al. (2018), Cinque (2023) and De Clercq et al. (2025).

3.3.6.2 Pronominal and auxiliary clitic clusters

Unlike the systems examined in the previous subsection, Slavic languages provide evidence from both pronominal and verbal clitics. The languages listed in Table 1.10 differ from those in Table 3.8 in that person features are expressed not only by pronominal clitics but also by verbal auxiliaries (Aux). Since the person-related positions are already occupied by verbal auxiliaries, pronominal clitics in Slavic are ordered predominantly according to case. Nevertheless, Person Case Constraint effects are occasionally attested; for discussion, see, for example, Stegovec (2015a,b) on Slovenian, Pancheva and Zubizarreta (2018) on Bulgarian, and Cetnarowska (2004) on Polish.⁷ With this background in mind, let us now turn to the data in Table 3.13, which illustrates the structure of clitic clusters in those Slavic languages that exhibit them.

Table 3.10: Clitic clusters ordered by person and case (pronominal and auxiliary clitics)

language (Indo-European, Slavic)	cluster
Serbian	C — Aux.T(1/2) — Dat — Gen/Acc — Refl — Aux.T(3)
Croatian	C — Aux.T(1/2) — Dat — Gen/Acc — Refl — Aux.T(3)
Slovenian	Aux.Cond — Aux.T(1/2) — Refl — Dat — Gen/Acc — Aux.T(3)
Bulgarian	Aux.Cond/Aux.T(1/2) — Dat — Acc — Aux.T(3)
Macedonian	Aux.Cond — Aux.T(1/2) — Dat — Acc — Refl/Aux.T(3)
Slovak	Aux.Cond — Aux.T(1/2) — Refl — Dat — Gen/Acc
Sorbian	C — Aux.T(1/2) — Refl — Dat — Acc
reference	Browne (1974), Dimitrova-Vulchanova and Hellan (1999) Milićević (1999), Franks and King (2000), and Bošković (2001)

In Croatian and Serbian, the clitic cluster follows a consistent structural pattern: it begins with the complementiser (C) followed by past tense auxiliaries marked for person (Aux.T(1/2)), then object clitics marked for case, and ends with auxiliaries marked for past tense but unmarked for person (Aux.T(3)). This ordering, shown in (24a), aligns with the general clausal hierarchy presented in (23). More specifically, person features shape the structure in the following way: the upper tense auxiliaries bear 1ST or 2ND person features (Aux.T(1/2)) which enables them to be at the

⁷In Table 3.13, we also include the position of reflexive clitics. As the reader will observe, their placement is not uniform across Slavic languages. The same phenomenon was attested in Romance languages. We take this to indicate that the reflexive clitic fulfills different functions across languages, which is reflected in its position within the cluster. We nevertheless include them in the clitic clusters for readers who may be interested in their placement. A full account of reflexive clitics lies beyond the scope of this chapter, however, Taraldsen Medová discusses this topic in another chapter of this book, see Chapter *The SE illusion: Argumental looks, antipassive reality*, Section 2.3.

higher position in the clitic cluster, while the final auxiliary is unmarked for person (Aux.T(3)), i.e., bears 3RD person, and thus stays in the lower position. This distribution respects the person hierarchy, as shown in (24b). Finally, the pronominal object clitics appearing between Aux.T(1/2) and Aux.T(3), including dative and genitive/accusative, are ordered according to the case hierarchy, shown in (24b).

- (24) a. CP > Mood > PersonP > CaseP > TP
 b. 1/2P > 3P
 c. DatP > Gen/AccP

In Slovenian, the same logic applies, with the conditional (Aux.Cond) and 1ST or 2ND person tense auxiliary at the left edge of the clitic cluster, and the 3RD person tense auxiliary at the right edge. Case clitics (dative and genitive/accusative) are located directly before the 3RD person auxiliary, see (25).

- (25) a. MoodP > PersonP > CaseP > TP
 b. 1/2P > 3P
 c. DatP > Gen/AccP

In Bulgarian and Macedonian, the ordering principle is similar to (25). The cluster starts with the conditional auxiliary, followed by the auxiliary with 1ST/2ND person features. Case clitics (dative and accusative) appear next, and the tense auxiliary unmarked for person (i.e., 3RD person) occupies the final position.

Table 3.13 illustrates the distribution of independent features clearly: tense auxiliaries appear in two different positions depending on the person features they bear. Auxiliaries with 1ST/2ND person features sit very high in the cluster, with only conditional or CP domain clitics above them, while auxiliaries with 3RD person features are realised at the right edge. Comparably with Romance and Bantu languages, the relative ordering between the person domain and the case domain remains stable: marked person features (1/2) precede case features, and the 3RD person auxiliary surfaces at the bottom of the case domain. The hierarchy PersonP > CaseP is thus confirmed.

Moreover, the position of the 3RD person auxiliary, AUX.T(3), shows where Tense is first realised in the sequence, namely below CaseP. This supports the placement that CaseP is above TP. At the same time, the data demonstrate that 1ST/2ND person is lower than mood and complementiser features, yielding the hierarchy in (26) as the overall ordering. And this is precisely the ordering we have got in (23).

- (26) CP > MoodP > 1/2P > DatP > AccP > TP > AspP

In Slovak, and Sorbian, the clitic cluster ordering follows the same pattern, with complementisers and auxiliaries at the left edge and case clitics at the right. What distinguishes these languages is that no 3RD person auxiliary occurs at the end of the cluster, so the sequence concludes with the case clitics. The question of why 3RD person auxiliaries are absent in some Slavic languages, and how this should be modeled, is left open. What matters here is that in other Slavic languages this

position is overtly realised, which provides evidence that it exists and plays a role in the structure of the clitic cluster.

Yet, the picture is not always as neat as the descriptions of the standard variants of Slavic languages might suggest. Data from non-standard varieties reveal patterns that are both interesting and challenging, and they occasionally appear to run against the generalisations established so far.

Similarly to the data in the previous sections, Slavic languages also have their own “trouble-makers” that at first sight seem to go against our generalisation. One such case is a non-standard variety of Slovenian, illustrated in Table 3.11. As shown there, clitics may violate the expected order Dat — Acc, since both Dat — Acc and Acc — Dat orders may occur.

Table 3.11: Clitic clusters ordered by person and case features (Slavic troublemaker)

reference	language	cluster
Stegovec (2015b)	Non-standard	1/2 — 3
	Slovenian	<i>and</i>
	(Indo-European)	3(Dat) — 3(Acc) or 3(Acc) — 3(Dat)

To address this, we return to the Greek and French examples, which showed similar behaviour in the very same morphosyntactic contexts. In non-standard Slovenian, just like in Greek or French, Acc — Dat arises mostly with non-finite verbs, especially imperatives (Stegovec, 2015b). If we apply the same logic to Slovenian as we did to Greek and French, the apparent exception disappears (see (20a), (20b), and (20c)).

Non-standard Slovenian is also exceptional in being sensitive to person (and showing PCC effects). As Table 3.13 has shown, Slavic pronominal clitics are generally ordered by case, with person irrelevant. However, non-standard Slovenian in Table 3.11 shows a tendency to reflect person conditioning (1/2 — 3). In this respect, it patterns more like the languages in Table 3.8 than like its closest Slavic relatives.

Seen in this light, non-standard Slovenian is not a real troublemaker but rather a connecting piece that strengthens the overall argument. Standard Slovenian patterns together with the other Slavic languages, exactly as expected, while the non-standard variety aligns with Romance, Bantu, and Greek. A language like Slovenian thus shows that even though clitic clusters may look different on the surface (depending on which subset of features is active) their ordering follows the same underlying principle. Within a cartographic approach, these rigid orderings correspond to the universal fseq, which is shared across languages, as the preceding discussion has tried to demonstrate.

What we have established on the basis of clitic clusters goes hand in hand with findings from independent sources: non-clitic data likewise point to the presence of dedicated person projections located on the borderline of CP and TP domains. Such ordering of projections has been proposed, for example, by Alexiadou and Anagnostopoulou (2006), Fábregas (2018), Cortiula (2023) and dos Santos (2025). These independent arguments therefore converge with the clitic data. Within the cartographic tradition on clitics, similar proposals have been advanced by Poletto (2000), Bianchi (2006) and Ciucivara (2009); Manzini and Savoia (2016) for Romance languages.

In the next section, we turn to examples from Italian and French, which have been argued to include both subject and object pronominal clitics. These data will allow us to support the placement of the subject projection above the person projection, and its nature will be further discussed based on Poletto (2000).

3.3.6.3 Pronominal clitic clusters with subject

There are not many languages whose clitic clusters include both subject and object clitics and are also ordered with respect to both case and person. Among those that do, French and Italian are particularly relevant. Both exhibit clitic clusters that contain subject and object clitics simultaneously (see Table 3.12).

Table 3.12: Clitic clusters ordered by person, case and grammatical function features

reference	language	(proclitic) cluster
Bonet (1991) Seuren (2009) Sportiche (2017)	French (Indo-European)	Subj — Obj <i>and</i> Obj[1/2 — 3(Dat) — 3(Acc)]
Monachesi (1997) Poletto (2000) Cardinaletti (2008)	Italian (Indo-European)	Subj — Obj <i>and</i> Obj[1/2 — 3 and Dat — Acc]

In both French and Italian, the proclitic cluster always begins with a subject clitic, followed by object clitics.⁸ The object clitics are ordered according to both case and person features. In French, the attested orderings show that 1ST and 2ND person object clitics precede 3RD person object clitics, and among 3RD person object clitics, datives precede accusatives. Italian exhibits the same pattern: subject clitic first, followed by object clitics ordered by person features (1ST/2ND > 3RD), and followed by ones ordered by case features (Dat > Acc).

The data in Table 3.12 show that clitics bearing a subject function (regardless of their person or case specification) consistently appear at the top of the clitic cluster, see (27).

(27) SubjP > Obj[Person > Case]P

This firstly supports the generalisation in (26), but it also raises an important question: what kind of feature ensures that the subject clitic always surfaces at the top of the cluster in French and Italian, if it is neither person nor case?

Poletto (2000), in her study of Italian dialects, observes that subject clitics vary not only in person, number, and gender, but also in features related to information structure. Specifically, some clitics encode distinctions between given and new information, referred to as theme and rheme. Based on this, Poletto proposes that subject clitics appear in the highest positions because

⁸This holds for any clitic cluster in a declarative clause that contains both subject and object clitics.

they spell out features from both the highest IP and CP domains. From the IP domain, these include number, person, and gender projections; from the CP domain, projections associated with information structure, such as given and new information.

On the basis of these observations, Poletto (2000) proposes a list of projections forming the higher functional field: see (28). Since her study focuses exclusively on different types of subject clitics (typically marked by nominative), case distinctions are not included. As a result, the position of case within the broader hierarchy cannot be established from her data. Nonetheless, the proposal strongly supports the view that subject clitics express CP domain features, structurally dominating person and case.

(28) CP > New/Given_InfoP > NumberP > PersonP > GenderP > TP

While Poletto's sequence includes additional features such as Number and Gender, these lie beyond the scope of the present study as we do not attempt to determine the relative position of number or gender features.

Thus, although Poletto's proposal includes Number and Gender projections, we do not examine these features further. Therefore, we simplify Poletto's hierarchy as follows, (29):

(29) CP > New/Given_InfoP > PersonP > TP

After simplification, our universal functional sequence includes two distinct positions in which, according to Poletto, subject clitics appear: a lower one that spells out person features, and a higher one that spells out features related to information structure. If we now combine Poletto's findings with our previous generalisations, we arrive at a more specified picture of the clitic zone. In earlier sections, we established that marked person features (1ST/2ND) precede case features (Dat > Acc) (30a), and that all grammatical functions, person and case are positioned between MoodP and TP (30b). Poletto's research adds a higher layer: features related to information structure, which she places above person. These features account for the initial surface position of subject clitics.

- (30) a. generally: CP > MoodP > Grammatical_functionP > TP > AspP
 b. in more detail: CP > MoodP > SubjP > ObjP > TP > AspP
 c. in more detail: CP > MoodP > SubjP > 1/2P > DatP > AccP > TP > AspP

The resulting functional sequence, integrating both sets of findings (ours and Poletto's), is given here:

(31) CP > MoodP > New/Given_InfoP > 1/2P > DatP > AccP > TP > AspP

This unified sequence preserves the universal verbal (clausal) hierarchy, establishes the internal structure of the person and case hierarchies as well as their mutual ordering, and includes the information-structural layer needed to explain the consistently high position of subject clitics. And now we compare this hierarchy with the clitic cluster in Czech.

3.3.7 Czech clitic cluster ordering

Having established the cross-linguistic generalisations emerging from clitic ordering, we now turn to Czech and ask whether its clitic cluster is compatible with the functional sequence proposed above. Czech is a particularly relevant testing ground in this respect, since it displays rigidly ordered clitic clusters which have been discussed extensively in the literature (references are provided in Table 3.13). The question is whether this ordering can be understood as another reflex of the universal hierarchy argued for in the preceding sections. At a descriptive level, the Czech cluster appears to conform to a stable ordering in which complementiser-related material is followed by conditional and auxiliary elements, then by dative clitics, and finally genitive/accusative⁹ clitics, with reflexive material occupying an intermediate position that will not be addressed here in detail. In the terms adopted here, this can be represented as in Table 3.13:

Table 3.13: Czech¹⁰

language	cluster
Czech	C > Aux.Cond > Aux.T(1/2) > Refl > Dat > Gen/Acc
reference	Zwicky (1977), Fried (1994), Avgustinova and Oliva (1995), Veselovská (1995), Toman (1996), Fried (1999), Franks and King (2000), Lenertová (2001), Lenertová (2004), Rezac (2005), Hana (2007), Tanenbaum (2020), Adam (2024), Marušič et al. (2024) and Tabachnick (2024)

Examples in (32) illustrate the stable ordering generalisations within the Czech clitic cluster. In (32a), the sequence *jsem se mu ho* shows that the 1st person element precedes the reflexive and both pronominal clitics, yielding (33a). In (32b), the sequence *by se mu ho* shows that the conditional element likewise precedes the reflexive and the pronominal clitics, giving (33b). In (32c), where both elements co-occur, their relative order is fixed: *by* precedes *jsem*, which yields (33c). The last two examples add the complementiser clitic *-li* and show that it occupies the leftmost position within the cluster. In (32d), *-li* precedes the 1st person element, giving (33d). In (32e), it precedes the conditional element, giving (33e). Taken together, these examples reveal partial hierarchies that converge on a consistent ordering of complementiser, person, conditional, reflexive, and case-marking pronominal clitics.

⁹The ordering of genitive clitics is less straightforward, since it depends on the function of the relevant pronominal clitic in a given configuration, including uses associated with genitive and accusative. In terms of ordering, object genitive does not behave in the same way as genitive assigned within a quantificational phrase. We therefore leave this issue open and label the relevant position Gen/Acc.

¹⁰Březina in Chapter ‘To be’ in the Old Czech cluster investigates Czech clitic cluster diachronically.

- (32) a. Já **(j)sem se mu ho** snažila představit.
 I.NOM AUX.T.1SG REFL him.DAT him.DAT him.ACC try.PST.F.SG introduce.INF
 'I tried to introduce him to you.' (Standard/Common Czech)
- b. Já **bych se mu ho** snažila představit.
 I.NOM AUX.COND.1SG REFL him.DAT him.ACC try.PST.F.SG introduce.INF
 'I would try to introduce him to you.' (Standard Czech)
- c. Já **by (j)sem se mu ho** snažila představit.
 I.NOM AUX.COND AUX.1SG REFL him.DAT him.ACC try.PST.F.SG introduce.INF
 'I would try to introduce him to you.' (Common Czech)
- d. Snažila-li **jsem se mu ho** představit, bylo to jen
 try.PST.F.SG-C AUX.T.1SG REFL him.DAT him.ACC introduce.INF be.PST.N.SG it.NOM only
 jednou.
 once
 'If I tried to introduce him to him, it was only once.' (Standard Czech)
- e. Snažila-li **bych se mu ho** představit, bylo by
 try.PST.F.SG-C COND.1SG REFL him.DAT him.ACC introduce.INF be.PST.N.SG AUX.COND
 to jen jednou.
 it.NOM only once
 'If I tried to introduce him to him, it would be only once.' (Standard Czech)
- (33) a. Aux.T(1) > Refl > Dat > Acc
 b. Aux.Cond(1) > Refl > Dat > Acc
 c. Aux.Cond > Aux.T(1) > Refl > Dat > Acc
 d. C > Aux.T(1) > Refl > Dat > Acc
 e. C > Aux.Cond(3) > Refl > Dat > Acc

In Czech, 1ST and 2ND auxiliary clitics occupy a high position within the clitic cluster, as they do in other Slavic languages, because they realise person features located in this area of the sequence. The only clitics that precede them are higher mood-related clitics, in Czech, the conditional ones, and complementiser clitics.¹¹ The lower part of the Czech cluster shows a stable ordering in which dative clitics precede genitive/accusative ones. Czech therefore matches the cross-linguistic sequence proposed above. With this, the Czech data support our final form of the universal hierarchy (31), discussed previously in this section.

3.3.8 Summary of the cross-linguistic patterns in clitic ordering

The point of departure for this study is the well-known gap in our understanding of the universal functional sequence. While CP and TP are well established, the projections intervening between

¹¹For further discussion of Czech tense auxiliary clitics, see the chapter by Martin Březina, *'To be' in the Old Czech cluster*, who examines them from a diachronic perspective. An analysis of Czech conditional clitics is provided in chapter *A mirror-principle violation meets Nanosyntax: The case of Czech X-marking* by Pavel Caha, where he offers a nanosyntactic analysis of the two conditional clitic forms *bychom* and *bysme* 'we would'.

them, where clitics typically appear, are much less clearly understood. This gap can be illustrated by the following scheme (34):

$$(34) \quad \text{CP} > \text{clitic zone} > \text{TP} \leftrightarrow \text{CP} > ??? > \text{TP}$$

Our approach has been to use clitic clusters as a diagnostic. Following the cartographic methodology, which treats rigid ordering as evidence for dedicated functional projections, we asked whether the internal order of clitics can reveal which features are actually projected in this part of the functional sequence. To this end, we surveyed clitic clusters from over 30 languages and their varieties and took them into account in our analysis.

The survey has begun with languages where clitic clusters reflect only verbal features (Datooga and Ihanzu–Nyilamba in Table 3.1). These systems, though relatively simple, already confirm that clitic ordering follows the clausal sequence in (35).

$$(35) \quad \text{CP} > \text{MoodP} > \text{TP} > \text{AspP}$$

Having established this baseline, we have then turned to languages in which clitics encode grammatical functions. In Iraqw–Gorwaa, Alagwa, and Nyaturu (Table in 3.2), subject clitics systematically precede object clitics, and both are placed above T, see (36):

$$(36) \quad \text{SubjP} > \text{ObjP} > \text{TP}$$

These data establish that ordering based on grammatical function is respected in the clitic zone.

Another piece of evidence has come from person-based clitic clusters (Table 3.4). In Atayal, Warlpiri, and Pashto, the cluster is ordered exclusively by person, with 1ST/2ND person clitics preceding 3RD ones. The person hierarchy in (37) is thus respected.

$$(37) \quad 1/2\text{P} > 3\text{P}$$

These languages confirm that person features are projected within the clitic zone and independently contribute to the internal structure of the clitic cluster.

Once both grammatical function features and person features are considered together, languages such as Wolof (Table 3.5) show that the two interact in a systematic way: subject clitics consistently precede object clitics, and among object clitics, the person hierarchy (1/2 > 3) is observed. This pattern supports a structure in which subject-related projections are located higher than projections associated with person-marked object clitics, see (38).

$$(38) \quad \text{SubjP} > \text{ObjP} \text{ and } \text{Subj} > 1/2\text{P} > 3\text{P}$$

Case features introduce yet another feature layer within the clitic zone. Languages such as Kapampangan and Mamanwa (Table 3.6) have provided evidence that case (as a morphosyntactic reflection of grammatical function) alone can order clitics showing the pattern, (39):

$$(39) \quad \begin{array}{ll} \text{a.} & \text{Gen(Erg)P} > \text{Nom(Abs)P} \\ \text{b.} & \text{MoodP} > \text{CaseP} > (\text{TP}) > \text{AspP and Dat} > \text{Acc} \end{array}$$

When we have considered the interaction between case features and verbal features, Kambera has provided crucial evidence: it shows that clitic case projections are positioned between the modal and aspectual projections, and that they respect the universal case hierarchy, with dative above accusative in (39b).

Together, these generalisations show that all three grammatical domains (grammatical function, person, and case) are directly responsible for clitic ordering and must be projected in a high domain, such as the clitic zone.

A fuller picture has emerged in Romance, Bantu, and Greek (Table 3.8). These languages show that two hierarchies jointly determine ordering of the pronominal clitic cluster: 1ST and 2ND person features are higher than case features, and case distinctions, (40a), are distinguished within 3RD person clitics, (40b,c):

- (40) a. Dat > Acc and 1/2P > 3P
 b. 1/2P > 3(Case)P
 c. 1/2P > DatP > AccP

These systems therefore reveal a stable interaction between person and case features and show that person-based projections are higher in the functional sequence than case-based ones.

Further, Slavic languages (Tables 3.13–3.11) broadened the picture by adding verbal auxiliaries into the cluster. Auxiliaries bearing 1ST/2ND person features occur high in the cluster, providing further evidence that person is projected in this position of the hierarchy. Moreover, this goes in line with Romance and Bantu data. Auxiliaries bearing 3RD person features appear at the lower edge of the case domain, marking the first realisation of Tense below CaseP. This confirms the full sequence in (41):

- (41) a. CP > MoodP > PersonP > CaseP > TP > AspP
 b. CP > MoodP > 1/2P > DatP > AccP > TP > AspP

Finally, we have shown that in French and Italian, subject clitics consistently appear above object clitics and are merged between mood and person projections. This position places subject clitics structurally higher than both person and case projections. Following Poletto (2000), we take this to indicate that certain subject clitics express features associated with information structure, such as the contrast between given and new information, see (42). We assume that this accounts for their consistently high placement in the clitic cluster.

- (42) a. CP > MoodP > SubjP > 1/2P > DatP > AccP > TP > AspP
 b. CP > MoodP > New/Given_InfoP > 1/2P > DatP > AccP > TP > AspP

Although every set of languages examined contains its own “troublemakers” – clusters that at first sight seem to contradict the ordering predicted by the fseq (i.e., Swahili, French, Greek, Barceloní Catalan, Romanian, or non-standard Slovenian) – their surface patterns can be captured by movement (or movement plus pied-piping). Crucially, such cases are rare. Precisely because movement only rarely interferes, clitic clusters provide especially clear and direct evidence for the ordering of individual projections within the universal functional sequence. In line with the cartographic

framework, rigid clitic ordering shows which features are projected in the clitic zone and how they are hierarchically ordered. At the same time, we have to frankly admit, that the list of languages presented here is not exhaustive. Further, we do not capture all the features that may be relevant (e.g., number or gender), and thus further research is needed to refine the picture.

3.4 Further perspectives

The central claim of this chapter is that the rigid ordering of clitics reflects the universal functional sequence (fseq). In other words, the order in which clitics appear in the clitic cluster is not arbitrary, but mirrors the hierarchical organisation of morphosyntactic features. This view requires a morphosyntactic model in which the presence of a given feature triggers the movement of material from a lower part of the structure into the clitic zone, where the relevant feature projection must be merged, as required by the fseq.

Such a model is provided by Starke (2004, 2024, 2025), who argues that merge and movement are triggered by the requirements of successful fseq-realisation. As Starke puts it, (43):

- (43) a. *There exists an ‘fseq’ – a sequence of functional projections – such that the output of merge must respect fseq, modulo unmarked values.* (Starke, 2004, p. 256)
 b. *[...] insertion is triggered directly by fseq. The fseq requirement itself thus triggers merge/move. [...] We thus get a trigger mechanism ‘for free’.* (Starke, 2004, p. 257)

Since we argue that the ordering of the clitic cluster is determined by the universal fseq, we eliminate from the analysis all redundant mechanisms that have so far been arbitrarily formulated for selected positions around CP and TP. Subsequently, we are left with a single simple rule: respect the fseq and help it realise itself when it needs to.

Therefore, whenever a clitic reflects a certain feature – such as Person, Case, Tense, or Mood – it appears in a specific position because the successful realisation of that feature requires syntax to create a projection at that point in the fseq. Some portion of the syntactic structure is then merged/moved into that position so that the feature can be realised. Since fseq is strictly ordered, the resulting clitic cluster is ordered strictly as well. Additional movements inside the clitic zone are rather rare, but not absent fully, as has been shown for Greek, Barceloní Catalan, Romanian, or non-standard Slovenian. Importantly, movement-based mechanisms developed in cartographic research are already part of Starke’s model within the so-called lexicalisation algorithm and are thus accounted for.

If we conclude, first, that the features involved in clitic cluster formation include person, case, aspect, tense, mood, information structure and complementiser features, and at the same time observe that their projections are rigidly ordered across a wide range of languages, then it follows that we can reconstruct the shape of the fseq on the basis of these features, in the clitic zone.

This ordering not only matches the verbal/clausal organisation (CP > MoodP > TP > AspP), but also incorporates nominal projections reflecting markedness scales: in the domain of grammatical functions SubjP > ObjP, in the domain of person features 1/2P > 3P, and in the domain of case DatP > AccP. Based on the observations presented in the previous section, we conclude that, at

least on the basis of the data examined here, the feature projections in the domain traditionally described as the clitic zone are ordered as shown in (44):

- (44) a. generally:
 CP > MoodP > New/Given_InfoP > PersonP > CaseP > TP > AspP
 b. in more detail:
 CP > MoodP > New/Given_InfoP > 1/2P > DatP > AccP > TP > AspP

Seen from this perspective, clitic clusters are not a deviant or exceptional phenomenon. They follow the same morphosyntactic principles that we already know from other non-clitic zones. What makes them valuable is precisely their transparency: because of their featural simplicity and rigid internal ordering, they reveal with particular clarity the position of the clitic zone within the universal fseq.¹²

It is precisely because the fseq is organised in this way that clitic ordering displays the patterns we find. In other words, the linear order of clitics directly mirrors the structure of the fseq: clitics appear in a given order because the fseq has that order, and the shape of the fseq can in turn be recovered from the ordering of clitics. **No additional ordering mechanism for ordering of clitics is therefore required.**

An approach of this kind is possible only in a theory that assigns this level of explanatory power to the fseq itself, for example the framework developed by Starke (2004), Caha (2009), Starke (2009, 2014), Taraldsen Medová and Wiland (2018) and De Clercq and Vanden Wyngaerd (2019). From this perspective, applying Nanosyntax to the derivation of clitics is a natural continuation of cartographic research.

3.5 Conclusion

We assumed that cross-linguistically fixed clitic ordering may serve as a diagnostic for identifying which morphosyntactic features are projected in the syntactic space between CP and TP. Adopting the cartographic One Feature One Head view, we treated the “clitic zone” not as a single head (one CIP) but as a field in which the feature projections realised by clitics, such as Complementiser, Mood, Information Structure, Person, Case, Tense and Aspect, are projected in a fixed order. The central aim was to determine the common underlying feature sequence directly from the linear order of clitics.

The survey of over 30 languages and their varieties supports this aim, since the observed clitic orders converge on a stable underlying feature sequence. Verbal-only clusters (Datooga, Ihanzu–Nyilamba) respect the basic clausal backbone CP > MoodP > TP > AspP. Languages without overt case marking (Iraqw–Gorwaa, Alagwa, Nyaturu) show that grammatical functions are projected, with subject clitics consistently preceding object clitics (SubjP > ObjP > TP). Case-only systems such as Kapampangan and Mamanwa demonstrate that clitics are ordered by case alone, following

¹²Caha in the first chapter *A mirror-principle violation meets Nanosyntax: The case of Czech X-marking*, thus treats clitic *by* as if it was a regular suffix of the verb when verb adjacent and in this manner the derivation of it proceeds. It is regular suffix derivation, no special treatment of clitics is needed.

the hierarchy Gen(Actor)P > Nom(Undergoer)P. Person-based systems such as Atayal, Warlpiri, and Pashto show that clitics marked for 1ST or 2ND person systematically precede those marked for 3RD person (1/2P > 3P). All these systems show, first, which features are projected in the clitic zone, and second, in what order they are projected. Further languages then show how these groups of features are ordered relative to one another.

Languages where person and case interact confirm and refine this picture. Wolof combines both principles: subjects precede objects, and among objects, the person hierarchy 1/2P > 3P determines their order. Romance languages reveal even more structure: 1ST and 2ND person clitics always precede 3RD person, and within the 3RD person domain, case ordering (DatP > AccP) applies. Greek, despite surface alternations, also reflects DatP > AccP in the underlying hierarchy. Slavic languages integrate both pronominal and verbal clitics: auxiliaries marked for 1ST/2ND person appear high in the sequence, followed by case clitics (DatP > AccP), while 3RD-person auxiliaries occur at the right edge, showing that Tense is realised below case. Finally, French and Italian proclitic clusters illustrate the structurally higher placement of subject clitics relative to other argument positions. These converging results yield the hierarchy established in this chapter, (45):

- (45) a. generally:
 CP > MoodP > New/Given_InfoP > PersonP > CaseP > TP > AspP
 b. in more detail:
 CP > MoodP > New/Given_InfoP > 1/2P > DatP > AccP > TP > AspP

The final section considered the proposal to apply Starke's theory to the derivation of clitics, since it offers a syntactic model in which the universal functional sequence plays a central role in determining the order of syntactic material. Clitic clusters across languages reveal a stable ordering in terms of feature representation, and this ordering closely matches the universal fseq discussed in theoretical work far beyond the domain of clitics. This makes it natural to adopt a model in which clitic ordering follows directly from the fseq. In Starke's approach, merge together with a small set of movement operations is sufficient to realise the universal sequence of projections, so clitic ordering emerges without any need for additional clitic-specific mechanisms such as special heads or morphological templates.

In sum, the data reveal both which features are ordered in the higher functional field between CP and TP and how they are ordered. The hierarchy in (45) is the structure uncovered by clitic clusters. For future research, this hierarchy forms the essential foundation from which clitic ordering can be derived, at least for the languages examined in this chapter. It provides a baseline, a universal fseq, that allows us to begin deriving clitic clusters.

It is important to recognise that establishing this baseline is only a small step toward solving the clitic problem, but it is a necessary one. Any generative morphosyntactic theory assumes that the output of the derivation depends on the universal fseq. If different researchers adopt different fseqs, they will necessarily reach different conclusions, because they begin from different baselines. Since any generative framework assumes a universal – and therefore, by definition, single – fseq, these baselines should not diverge. When they do, the resulting analyses become mutually incompatible

and cannot meaningfully inform one another. A shared sequence of features is therefore essential for developing compatible analyses of clitic clusters.

In this chapter, we have aimed to identify which feature orderings within the clitic cluster – and thus within the universal fseq – are shared across existing syntactic analyses of clitics. In doing so, we establish a common baseline from which further research can proceed.

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